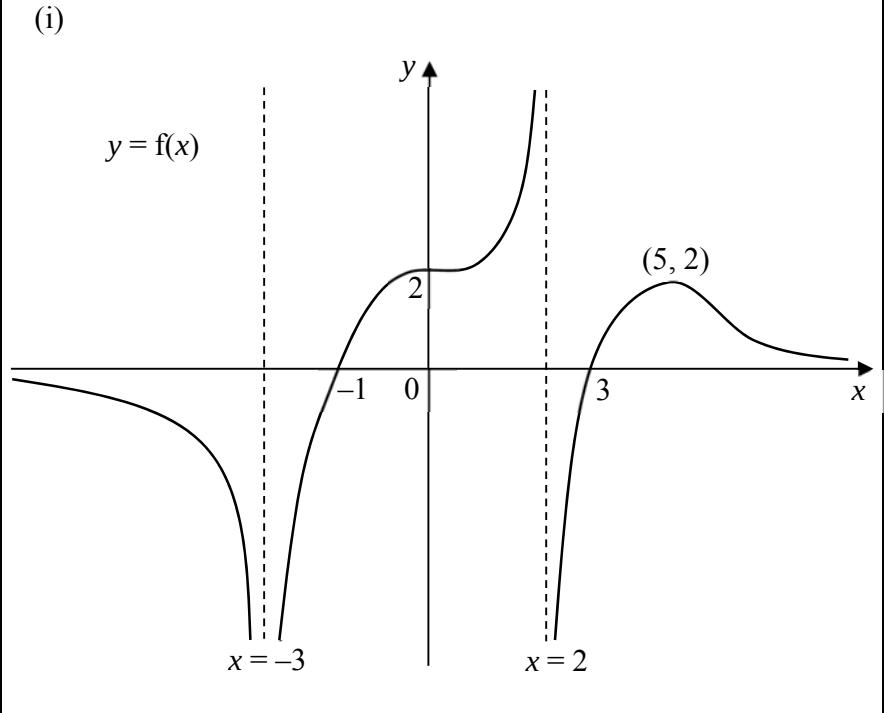
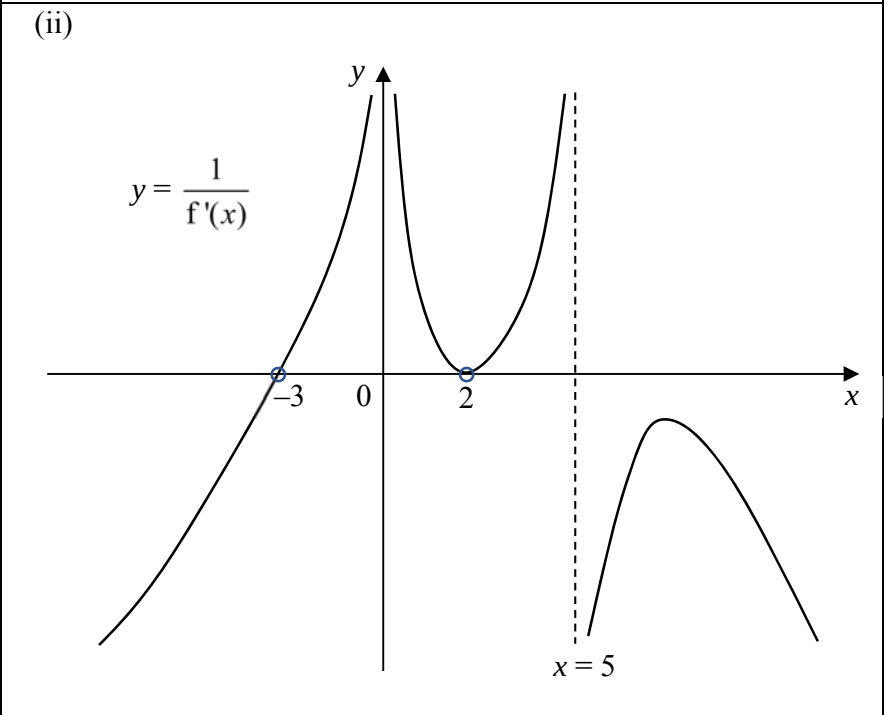


Solutions to 2018 Y6 H2 Maths Prelim Exam P1

1.	Solution [6 marks]	
	Given that $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are perpendicular, $\begin{pmatrix} p \\ q \\ r \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 0$ $\Rightarrow p + 2q + r = 0$ ---- (1) Let $\mathbf{n} = \overrightarrow{OA} \times \overrightarrow{OC} = \begin{pmatrix} p \\ q \\ r \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ $\mathbf{n} = \begin{pmatrix} p \\ q \\ r \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -r \\ r \\ p-q \end{pmatrix} = k \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ ---- (2) From (2): $\begin{cases} -r = -k \\ r = k \\ p - q = 2k \end{cases} \Rightarrow p - q - 2r = 0$ ---- (3) Let l be the line passing through the point D parallel to the vector $\overrightarrow{OB}$. $l: \mathbf{r} = \begin{pmatrix} 0 \\ -3 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ where $\beta \in \mathbb{R}$ Since A lies on l, $\begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ for some $\beta \in \mathbb{R}$ ---- (4) From (4): $\begin{cases} p = \beta \\ q = -3 + 2\beta \\ r = \beta \end{cases} \Rightarrow 2p - q = 3 \text{ or } 2r - q = 3$ ---- (5) Solving (1), (3) and (5), $p = 1$, $q = -1$, $r = 1$.	

2.	Solution [6 marks]
	(i) 
	(ii) 

3.	Solution [7 marks]	
	(i) $E = \frac{0.9qQ}{r}$ diff wrt t, $\frac{dE}{dt} = \frac{-0.9qQ}{r^2} \frac{dr}{dt}$	
	(ii) $0.2 \leq r \leq 1.4$ (since $-1 \leq \cos\left(\frac{1}{2}t\right) \leq 1$)	
	(iii) $r = 0.8 + 0.6 \cos\left(\frac{1}{2}t\right)$ Given: $Q = 4.0$, $q = 1.5$, $r = 1.1$. Subst $r = 1.1$ into eqn: $1.1 = 0.8 + 0.6 \cos\left(\frac{1}{2}t\right)$ $\cos\left(\frac{1}{2}t\right) = \frac{1}{2}$ $\frac{1}{2}t = \frac{\pi}{3} + 2k\pi \quad \text{or} \quad \frac{1}{2}t = \frac{5\pi}{3} + 2k\pi, \text{ for } k = 0, 1, 2, 3, \dots$ $\therefore t = \frac{2\pi}{3} + 4k\pi \quad \text{or} \quad \frac{10\pi}{3} + 4k\pi$ $\frac{dr}{dt} = -0.3 \sin\left(\frac{1}{2}t\right)$ At the instant where the particle is 1.1 units from the conductor, $\sin\left(\frac{1}{2}t\right) = \frac{\sqrt{3}}{2} \quad \text{or} \quad -\frac{\sqrt{3}}{2}$ $\therefore \frac{dE}{dt} = \frac{-0.9qQ}{r^2} \frac{dr}{dt}$ $= \left(\frac{-0.9(1.5)(4.0)}{1.1^2} \right) \left(-0.3 \frac{\sqrt{3}}{2} \right) \quad \text{or} \quad \left(\frac{-0.9(1.5)(4.0)}{1.1^2} \right) \left(0.3 \frac{\sqrt{3}}{2} \right)$ $= \pm 1.15947$ $= \pm 1.16 \text{ (to 3sf)}$	

4.	Solution [8 marks]	
	(i)(a) Note that origin is on plane Π. When O, A, B and C are coplanar, the line l lies on the plane Π, therefore the line l and plane Π intersect along the line l. (Points of intersection are points on l.) (i)(b) When O, A, B and C are not coplanar the line l and plane Π intersect at the point A.	
	(ii) $\frac{1}{2} \mathbf{a} \times \mathbf{b}$ gives the area of a triangle formed with two sides parallel and equal in magnitude to $\mathbf{a}$ and $\mathbf{b}$.	
	(iii) $\mathbf{c} \cdot \mathbf{p}$ gives the length of projection of $\mathbf{c}$ on $\mathbf{p}$ (<u>or</u> on $\mathbf{a} \times \mathbf{b}$, <u>or</u> on normal of Π).	
	(iv) Area of triangle OAB $= \frac{1}{2} \left \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right$ $= \frac{1}{2} \left \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right$ $= \frac{\sqrt{3}}{2} \text{ units}^2$ Using (iii) result, height of pyramid = height of point C relative to plane OAB = $\mathbf{c} \cdot \mathbf{p}$ where $\mathbf{p}$ is a unit vector in the direction of $\mathbf{a} \times \mathbf{b}$. $= \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \frac{2}{\sqrt{3}}$ Volume of the pyramid $OABC = \frac{1}{3} \times \frac{\sqrt{3}}{2} \times \frac{2}{\sqrt{3}} = \frac{1}{3} \text{ unit}^3$	

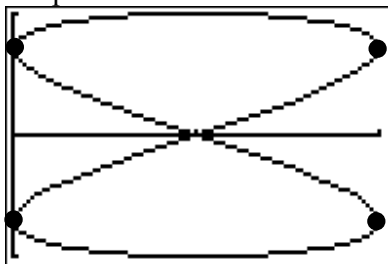
5.	Solution [8 marks]	
	(i) Note that $x = 2 \sin^2 \theta$ $\Rightarrow \frac{dx}{d\theta} = 2(2 \sin \theta) \cos \theta = 4 \sin \theta \cos \theta$. Also, when $x = 0$, $2 \sin^2 \theta = 0 \Rightarrow \theta = 0$; when $x = 1$, $2 \sin^2 \theta = 1 \Rightarrow \sin \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$ $\therefore \int_0^1 \sqrt{2x - x^2} \, dx$ $= \int_0^{\frac{\pi}{4}} \sqrt{4 \sin^2 \theta - (2 \sin^2 \theta)^2} \times 4 \sin \theta \cos \theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} \sqrt{4 \sin^2 \theta - 4 \sin^4 \theta} \times 2 \sin 2\theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} \sqrt{4 \sin^2 \theta (1 - \sin^2 \theta)} \times 2 \sin 2\theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} \sqrt{4 \sin^2 \theta \cos^2 \theta} \times 2 \sin 2\theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} 2 \sin \theta \cos \theta \times 2 \sin 2\theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} 2 \sin^2 2\theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} 1 - \cos 4\theta \, d\theta$ $= \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\frac{\pi}{4}}$ $= \frac{\pi}{4} - 0 = \frac{\pi}{4}$	
	(ii) $\int_0^1 \frac{x(1-x)}{\sqrt{2x-x^2}} \, dx = \frac{1}{2} \int_0^1 \frac{x(2-2x)}{\sqrt{2x-x^2}} \, dx$ $= \frac{1}{2} \int_0^1 (x) \times \frac{(2-2x)}{\sqrt{2x-x^2}} \, dx$ $= \frac{1}{2} \left[\int_0^1 x \, g(2\sqrt{2x-x^2}) \, dx - \int_0^1 2\sqrt{2x-x^2} \, g(1) \, dx \right]$ $= \frac{1}{2} \left[2 - 0 - 2 \int_0^1 \sqrt{2x-x^2} \, dx \right]$ $= \frac{1}{2} \left[2 - 2 \times \frac{\pi}{4} \right] = 1 - \frac{\pi}{4}$	

	<u>Alternatively,</u> $\int_0^1 \frac{x(1-x)}{\sqrt{2x-x^2}} dx = \int_0^1 \sqrt{2x-x^2} - \frac{x}{\sqrt{2x-x^2}} dx$ $= \int_0^1 \sqrt{2x-x^2} dx + \frac{1}{2} \int_0^1 \frac{-2x+2}{\sqrt{2x-x^2}} - \frac{2}{\sqrt{2x-x^2}} dx$ $= \frac{\pi}{4} + \frac{1}{2} \int_0^1 \frac{-2x+2}{\sqrt{2x-x^2}} dx - \int_0^1 \frac{1}{\sqrt{1-(x-1)^2}} dx$ $= \frac{\pi}{4} + \frac{1}{2} \left[2\sqrt{2x-x^2} \right]_0^1 - \left[\sin^{-1}(x-1) \right]_0^1$ $= \frac{\pi}{4} + \frac{1}{2} (2) - (\sin^{-1} 0 - \sin^{-1}(-1))$ $= \frac{\pi}{4} + 1 - \left(0 - \left(-\frac{\pi}{2} \right) \right)$ $= 1 - \frac{\pi}{4}$	
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6.	Solution [9 marks]	
	(i) $x = 1 - \sin 2t$ and $y = \cos t$ $\frac{dx}{dt} = -2 \cos 2t$ and $\frac{dy}{dt} = -\sin t$ $\frac{dy}{dx} = \frac{\sin t}{2 \cos 2t}$	
	(ii) For stationary points, let $\frac{dy}{dx} = 0$ $\sin t = 0$ $t = 0, \pi$ <i>correspondingly</i> $x = 1, 1$ $y = 1, -1$ Thus the two stationary points are (1,1) and (1, -1).	

(iii)

Graph of C



Normal at point P is parallel to x-axis implies tangent at point P is vertical (grad is undefined).

Set denominator of $\frac{dy}{dx}$ to be 0

$$\cos 2t = 0$$

$$2t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \text{ or } \frac{7\pi}{2}$$

$$t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4} \text{ or } \frac{7\pi}{4}$$

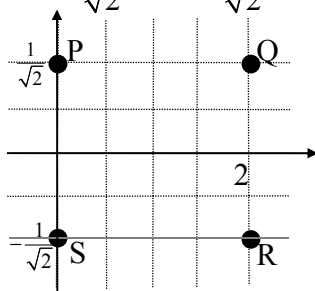
respectively,

$$x = 0, 2, 0 \text{ or } 2$$

$$y = \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \text{ or } \frac{1}{\sqrt{2}}$$

Thus the normal occur at four points

$$P(0, \frac{1}{\sqrt{2}}), Q(2, \frac{1}{\sqrt{2}}), R(2, -\frac{1}{\sqrt{2}}) \text{ or } S(0, -\frac{1}{\sqrt{2}})$$



The points form a rectangle with length 2 and breadth $\frac{2}{\sqrt{2}}$.

7.	Solution [9 marks]
	(i) $\sqrt{1-x^2} \frac{dy}{dx} = \frac{y}{2} \text{-----(1)}$ Diff Implicitly wrt x: $\sqrt{1-x^2} \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right) \left(\frac{1}{2}(1-x^2)^{-\frac{1}{2}}\right) (-2x) = \frac{1}{2} \frac{dy}{dx}$ $\sqrt{1-x^2} \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right) \left(\frac{-x}{\sqrt{1-x^2}}\right) = \frac{1}{2} \frac{dy}{dx} \text{-----(2)}$ $\times \sqrt{1-x^2}$ throughout, $(1-x^2) \frac{d^2y}{dx^2} - x \left(\frac{dy}{dx}\right) = \frac{1}{2} \frac{dy}{dx} \sqrt{1-x^2}$ $(1-x^2) \frac{d^2y}{dx^2} - x \left(\frac{dy}{dx}\right) = \frac{y}{4} \text{ (from (1))}$ Diff Implicitly wrt x: $(1-x^2) \frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} (-2x) - \left[x \frac{d^2y}{dx^2} + \frac{dy}{dx} \right] = \frac{1}{4} \frac{dy}{dx}$ $(1-x^2) \frac{d^3y}{dx^3} - 3x \frac{d^2y}{dx^2} = \frac{5}{4} \frac{dy}{dx} \text{-----(3)}$ Given that $x = 0, y = 1$, From (1), (2) and (3) respectively $\frac{dy}{dx} = \frac{1}{2}, \quad \frac{d^2y}{dx^2} = \frac{1}{4}, \quad \frac{d^3y}{dx^3} = \frac{5}{8}$ $\therefore y = 1 + x\left(\frac{1}{2}\right) + \frac{x^2}{2!}\left(\frac{1}{4}\right) + \frac{x^3}{3!}\left(\frac{5}{8}\right) + \dots$ $= 1 + \frac{x}{2} + \frac{x^2}{8} + \frac{5x^3}{48} + \dots \text{(up to the } x^3 \text{ term)}$ <u>Alternatively:</u> $(1-x^2) \left(\frac{dy}{dx}\right)^2 = \frac{y^2}{4}$ $(1-x^2) 2 \left(\frac{dy}{dx}\right) \left(\frac{d^2y}{dx^2}\right) - 2x \left(\frac{dy}{dx}\right)^2 = \frac{y}{2} \frac{dy}{dx}$ $4(1-x^2) \frac{d^2y}{dx^2} - 4x \frac{dy}{dx} = y$ $4(1-x^2) \frac{d^3y}{dx^3} - 8x \frac{d^2y}{dx^2} - 4x \frac{d^2y}{dx^2} - 4 \frac{dy}{dx} = \frac{dy}{dx}$ $4(1-x^2) \frac{d^3y}{dx^3} - 12x \frac{d^2y}{dx^2} - 5 \frac{dy}{dx} = 0 \dots \dots \text{(the rest is the same as abv)}$

	(ii) From (i)'s answer, diff term by term $y' = \frac{dy}{dx} = \frac{1}{2} + \frac{x}{4} + \frac{5x^2}{16} + \dots \text{(up to the } x^2 \text{ term)}$ Using the standard Maclaurin's formula for e^x, $e^{y'} \approx e^{\frac{1}{2} + \frac{x}{4} + \frac{5x^2}{16}}$ $= e^{\frac{1}{2}} e^{\frac{x}{4} + \frac{5x^2}{16}}$ $\approx e^{\frac{1}{2}} \left(1 + \frac{x}{4} \right) \quad \text{(up to the } x \text{ term)}$ Hence $p = \sqrt{e}$ and $q = \frac{\sqrt{e}}{4}$	
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8.	Solution [11 marks]	
	(a)(i) $z^* = 4$ since $z = 4$ $\left \frac{(1+ai)^3}{(1-i)^2} \right = 4$ $\frac{ (1+ai) ^3}{ (1-i) ^2} = 4$ $\frac{(\sqrt{1+a^2})^3}{(\sqrt{2})^2} = 4$ $(\sqrt{1+a^2})^3 = 8$ $\sqrt{1+a^2} = 2$ $1+a^2 = 4$ $a = \pm\sqrt{3}$ $a = \sqrt{3} \text{ (reject -ve as } a \text{ is given to be positive)}$	

	(a)(ii) $\arg z^* = \arg \frac{(1+ai)^3}{(1-i)^2}$ $= 3 \arg(1+\sqrt{3}i) - 2 \arg(1-i)$ $= 3\left(\frac{\pi}{3}\right) - 2\left(-\frac{\pi}{4}\right)$ $= \frac{3\pi}{2}$ $= -\frac{\pi}{2} \text{ (principal value)}$ $\arg z^* = -\arg z$ $\therefore \arg z = \frac{\pi}{2} \text{ (principal value)}$ hence $z = 4e^{\frac{i\pi}{2}}$	
	(b) (i) since $i\pi$ is a root, $(i\pi)^4 + 2(i\pi)^3 + (\pi^2 + e^2)(i\pi)^2 + p(i\pi) + q = 0$ $\pi^4 + 2\pi^3(-i) + (\pi^2 + e^2)(-\pi^2) + i(p\pi) + q = 0$ $(q - e^2\pi^2) + i(p\pi - 2\pi^3) = 0$ Equating real and imaginary parts, $q = e^2\pi^2 \quad \text{and} \quad p = 2\pi^2$	

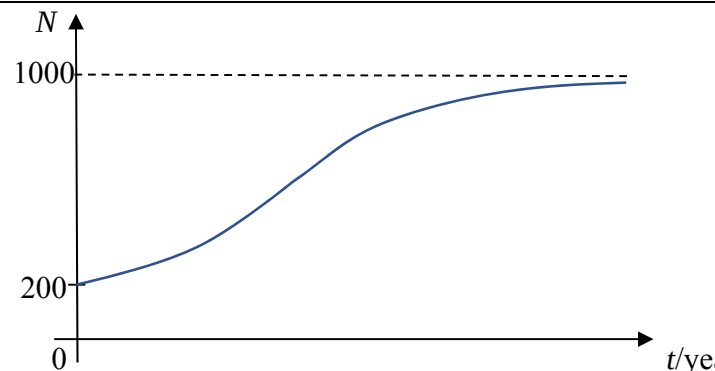
	(b)(ii) Now, $f(z) = z^4 + 2z^3 + (\pi^2 + e^2)z^2 + 2\pi^2z + e^2\pi^2$ As the polynomial $f(z)$ has real coefficients, $-i\pi$ is also a root. Thus $(z - i\pi)$ and $(z + i\pi)$ are factors of $f(z)$ i.e., $z^2 + \pi^2$ is a factor $f(z) = (z^2 + \pi^2)(z^2 + mz + e^2)$ where m is to be determined Equating coeff of z^3 or z in $f(z)$, $m = 2$ $f(z) = (z^2 + \pi^2)(z^2 + 2z + e^2)$ for $f(z) = 0$ $(z^2 + \pi^2)(z^2 + 2z + e^2) = 0$ $z = \pm i\pi \quad \text{or} \quad z = \frac{-2 \pm \sqrt{4 - 4e^2}}{2}$ $= -1 \pm \sqrt{1 - e^2}$ $= -1 \pm i\sqrt{e^2 - 1}$ (NOTE : $1 - e^2$ is negative) Hence the remaining roots are $-i\pi, -1 + i\sqrt{e^2 - 1} \quad \text{or} \quad -1 - i\sqrt{e^2 - 1}$	
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9.	Solution [11 marks]	
	(i) Let y_3, y_4 and y_6 be the 3rd, 4th and 6th terms of the arithmetic series. $y_3 = a, y_4 = ar, y_6 = ar^2$ $y_6 - y_4 = 2d = 2(y_4 - y_3)$ $\therefore ar^2 - ar = 2(ar - a)$ $r^2 - 3r + 2 = 0 \quad (\text{shown})$ $r = 2 \text{ or } r = 1 \text{ Reject since } r = 1 \text{ implies } d = 0.$	

	(ii) $\frac{y_4}{y_3} = r = 2$ $\Rightarrow y_4 = 2y_3$ $\Rightarrow a + 3d = 2(a + 2d)$ $\Rightarrow a = -d \text{ ---- (1)}$ $S_4 = -6$ $\frac{4}{2}[2a + (4-1)d] = -6 \text{ ---- (2)}$ Sub $a = -d$ into (2): $2d = -6$ $d = -3 \text{ and } a = 3$ <u>Alternatively</u> $\frac{y_4}{y_3} = \frac{y_6}{y_4} \Rightarrow \frac{a+3d}{a+2d} = \frac{a+5d}{a+3d}$ $\Rightarrow \dots\dots d(a+d) = 0$ i.e. $d = 0$ (n.a.) or $d = -a \text{ ---- (1)}$ Solving (1) and (2), $d = -3 \text{ and } a = 3$	
	(iii) Let g_m and a_m denote the m^{th} terms of G and A respectively. $ g_m - a_m > 10000 \text{ ---- (3)}$ Since $A: 3, 0, -3, -6, \dots$ while $G: -3, -6, -12, -24, \dots$, $a_m > g_m$ for all $m \in \mathbb{C}^+$. $ g_m - a_m > 10000$ $a_m - g_m > 10000$ $[3 + (m-1)(-3)] - (-3)(2)^{m-1} > 10000$ Using GC, when $m = 12$, $a_m - g_m = 6114 < 10000$ when $m = 13$, $a_m - g_m = 12255 > 10000$ Hence, the least value of $m = 13$.	

	(iv) $a_n = 3 + (n-1)(-3)$ $x_n = (2k)^{a_n}$ $x_n = (2k)^{3+(n-1)(-3)}$ $x_n = (2k)^3 \left[(2k)^{-3} \right]^{(n-1)}$ is a GP with first term $(2k)^3$ and common ratio $(2k)^{-3}$. The series $\sum_{n=1}^{\infty} x_n$ is the sum to infinity of the GP and it converges when $\left (2k)^{-3} \right < 1$ $\left \frac{1}{(2k)^3} \right < 1$ $(2k)^3 > 1 \text{ or } (2k)^3 < -1$ $2k > 1 \text{ or } 2k < -1$ $k > \frac{1}{2} \text{ or } k < -\frac{1}{2}$	
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10.	Solution [12 marks]	
	(i) Based on the proposed growth model by the scientists, $\frac{dN}{dt} = kN(1000 - N), \quad k > 0.$ Given that $\frac{dN}{dt} = 125$ when $N = 500$, $125 = k(500)(1000 - 500)$ $\Rightarrow k = \frac{125}{500 \times 500} = \frac{1}{2000}$ Thus, the required differential equation is $\frac{dN}{dt} = \frac{1}{2000} N(1000 - N)$	

(ii) Solving the D.E. $\frac{dN}{dt} = \frac{1}{2000} N(1000 - N)$, we have $\int \frac{1}{N(1000 - N)} dN = \int \frac{1}{2000} dt$ $\frac{1}{1000} \int \frac{1}{N} + \frac{1}{1000 - N} dN = \int \frac{1}{2000} dt$ $\int \frac{1}{N} + \frac{1}{1000 - N} dN = \int \frac{1000}{2000} dt = \int \frac{1}{2} dt$ Then, $\ln N - \ln 1000 - N = \frac{1}{2}t + c$ Simplifying, $\ln \left \frac{N}{1000 - N} \right = \frac{1}{2}t + c$ $\Rightarrow \left \frac{N}{1000 - N} \right = e^{\frac{1}{2}t + c} \Rightarrow \frac{N}{1000 - N} = \pm e^{\frac{1}{2}t + c}$ $\Rightarrow \frac{N}{1000 - N} = Ae^{\frac{1}{2}t} \quad \text{where } A = \pm e^c$ Given that when $t = 0$, $N = 200$, we have $A = \frac{200}{1000 - 200} = \frac{1}{4}$ Then, $\frac{N}{1000 - N} = \frac{1}{4}e^{\frac{1}{2}t}$ Making N the subject: $4N = (1000 - N)e^{\frac{1}{2}t}$ $\Rightarrow (4 + e^{\frac{1}{2}t})N = 1000$ $\Rightarrow N = \frac{1000e^{\frac{1}{2}t}}{4 + e^{\frac{1}{2}t}} = \frac{1000}{4e^{-\frac{1}{2}t} + 1}$	
	

	(iii) Based on the graph, we can deduce that the value of N tends towards 1000 as t tends to infinity. Thus, it is not possible for the scientist to achieve their target of exceeding 1200 plants in the area after 5 years.	
	(iv) (1) With severe climate changes in the wildlife area, there may be harsh conditions to deter growth of the plant and thus causing decline in their numbers at times; or (2) With the news of their cure for terminal illnesses, it may attract further unwelcomed visitors to the area for harvesting of the plant for private profits, thus again causing significant decline in their numbers.	

11.	Solution [13 marks]	
	(i) For the x-coordinates of intersection points P and Q, we $\text{let } \frac{1}{1+4x^2} = \frac{1}{\sqrt{5-4x^2}}$ Using GC, we have $x = \frac{1}{2} \text{ for point } P \text{ and } x = -\frac{1}{2} \text{ for point } Q$ The area of region R $= \int_{-1/2}^{1/2} \frac{1}{1+4x^2} - \frac{1}{\sqrt{5-4x^2}} dx$ $= \frac{1}{2} \int_{-1/2}^{1/2} \frac{2}{1+(2x)^2} dx - \frac{1}{2} \int_{-1/2}^{1/2} \frac{2}{\sqrt{5-(2x)^2}} dx$ $= \frac{1}{2} \left[\tan^{-1}(2x) \right]_{-1/2}^{1/2} - \frac{1}{2} \left[\sin^{-1} \left(\frac{2x}{\sqrt{5}} \right) \right]_{-1/2}^{1/2}$ $= \frac{1}{2} \left[\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right] - \frac{1}{2} \left[\sin^{-1} \frac{1}{\sqrt{5}} - \sin^{-1} \left(-\frac{1}{\sqrt{5}} \right) \right]$ $= \frac{\pi}{4} - \sin^{-1} \left(\frac{1}{\sqrt{5}} \right).$	

	(ii)(a) $y = \frac{1}{4x^2 - 8x + 5} = \frac{1}{4(x-1)^2 + 1}$ and $y = \frac{1}{\sqrt{1+8x-4x^2}} = \frac{1}{\sqrt{5-4(x-1)^2}}$ Thus, the original curves are translated 1 unit in the positive x-direction. Thus, the exact area is still $\frac{\pi}{4} - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ unit². (ii)(b) $y = \frac{2}{1+4x^2} = 2\left(\frac{1}{1+4x^2}\right)$, and $y = \frac{1}{\sqrt{\frac{5}{4} - x^2}} = \frac{1}{\sqrt{\frac{5-4x^2}{4}}} = \frac{2}{\sqrt{5-4x^2}} = 2\left(\frac{1}{\sqrt{5-4x^2}}\right)$ Thus, the original curves undergoes scaling by scale factor 2 parallel to the y-axis. The area of bounded region $= 2 \times \int_{-1/2}^{1/2} \frac{1}{1+4x^2} - \frac{1}{\sqrt{5-4x^2}} dx$ $= \frac{\pi}{2} - 2\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) \text{ unit}^2.$	
	(iii) For $y = \frac{1}{\sqrt{5-4x^2}}$: when $x = 0$, $y = \frac{1}{\sqrt{5}}$, and $x^2 = \frac{5}{4} - \frac{1}{4y^2}$ For $y = \frac{1}{1+4x^2}$: when $x = 0$, $y = 1$, and $x^2 = \frac{1}{4y} - \frac{1}{4}$ Also, $Q = \left(\frac{1}{2}, \frac{1}{2}\right)$	

	Then the required volume $= \pi \int_{\frac{1}{\sqrt{5}}}^{\frac{1}{2}} \left(\frac{5}{4} - \frac{1}{4} y^{-2} \right) dy + \pi \int_{\frac{1}{2}}^1 \left(\frac{1}{4y} - \frac{1}{4} \right) dy$ $= 0.174 \text{ unit}^3$	
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